

Program Name: Bachelor of Computer Application – Online
Semester: 3
1. Prerequisite: Object Oriented Programming

2. Rationale: Students will learn Software Application and Web Application development with C# language and MVC Architecture.

3. Detailed Syllabus:

Sr No.	Topics	Weightage (%)
1	Overview of Dot Net Framework, C# and ADO.Net: .Net Architecture and its components, Revising OOP Concepts with C# Language. Architecture of ADO.NET: Connected Architecture & DisConnected Architecture. ADO.NET Classes & Objects: Connection, Command, DataReader, DataAdapter, DataSet, DataColumn, DataRow, DataConstraints, DataView, Gridview Control, Repeater Control, Binding Data to DataBound Controls, SQLDataSource Control, DataBinding Expressions.	13
2	Overview of Windows Forms: Introduction to Windows Forms, Regular Controls in Windows Forms, Forms Architecture, Data Controls in windows forms, Data Binding with controls, Report Designing, Report Generation, Application Deployment, Working with Database for and Datasets.	15
3	Overview of ASP.NET Framework: Introduction Overview of Asp.NET Framework, Client Server Architecture, Application Web Servers, Installation of IIS server, Types of Files in ASP.NET, Types of controls in ASP.NET, Page Architecture, Page Class, Introduction to standard Controls: Buttons, Textbox, Checkbox, Label, Panel, Listbox, Dropdownlist, File Upload Control, Creatng Asp.Net Application & Adding Controls to a Webpage, Running an ASP.Net Application.	18
4	ASP.NET Validation Controls: Introduction to Validation, Client-Side Validation, Server-Side Validation. Types of Control: RequiredField Validator, Range Validator, CompareField Validator, RegularExpression Validator, Custom Validator, ValidationSummary Control.	12
5	State Management: Introduction to State, Requirement of State in Asp.Net, Client-Side State Management, Server-Side State Management, Various State Management Techniques: View State, Query String, Cookie, Session State, Application State.	12
6	Master page: Overview of Master Page, Requirement of a Master Page in an Asp.NET application, Designing Website with Master Page and customized CSS.	10
7	ASP.NET Application Configuration and Deployment of Application: Introduction To Web.Config, Common Configuration Sections, AppSettings, Tracing, Custom Errors, Authentication and Authorization, Deployment of Application in web server.	10

4. Reference Books:

1. **Asp.Net 4.0' Unleashed**
By Kevin Hoffman, Stephen Walther | 201
2. **Beginning ASP.NET4.5 in C.Sharp**
By Mathew Mac Donald | Third Edition

5. Course Outcome:

After the completion of the course, the students will be able to:

- 1.Understand .NET framework architecture and its features.
- 2.Build dynamic web applications to demonstrate event handling, state management and content manipulation using C#.
- 3.Develop data driven applications using ADO.NET, LINQ, XML and web service.

4. Practical List

Sr No.	Practical List
1	CRUD Operation with C# language and ADO.Net
2	Application Development with C# language and ADO.Net
3	Software Development using Windows Forms and ADO.Net
4	Develop a windows form is used for displaying the following details: 1. Student Name 2. Student ID 3. Student Location 4. Age 5. Gender Using data Concept of Data Binding in Asp.Net.
5	Design an ASP.NET Webpage to demonstrate the Label, Button and Textbox controls.
6	Design an Asp.NET Webpage that contain Listbox & Dropdownlist.
7	Develop a Registration Form with all Validation Controls.
8	Create a Web Service for all Arithmetic operations.
9	Develop a State Management in ASP.NET used in password authentication and user authorization through the session & Cookie.
10	Create a Master page using ASP.NET.
11	Full Web application development using ASP.Net